

Validity Analysis: ARABICMMLU

Target context: Non-Arab Tourists and Expats in Eight Arabic-Speaking Countries (ArabicMMLU Deployment)

Overall risk: **HIGH**

Deployment Context

Use case: Support foreigners visiting Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA in answering their general questions

Target population: Non-Arab tourists or expats visiting Morocco, Egypt, Jordan, Palestine, Lebanon, UAE, Kuwait, or KSA and have general questions about the Arabic language, history or geography of these countries

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content △	2	Significant gaps	high
Input Form ✓	3	Moderate gaps	high
Output Ontology △	1	Serious concern	high
Output Content △	2	Significant gaps	high
Output Form △	1	Serious concern	high
Average	2.0		

△ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

ArabicMMLU is a serious ground-up Arabic-language knowledge benchmark whose subject taxonomy and authentic source materials provide reasonable construct validity for evaluating school-curriculum knowledge of Arabic language, history, and geography. For the specific deployment of non-Arab tourists/expats across eight Arabic-speaking countries seeking educational-level knowledge, however, two structural mismatches dominate the assessment: (1) the single-correct-answer MCQ output ontology and form fundamentally cannot evaluate the multi-perspective, explanatory natural-language responses the deployment explicitly requires [Q27, Q88, elicitation A3]; and (2) input content and verification skew heavily Jordanian, with Morocco's school-curriculum content largely absent, Kuwait effectively absent, and no Palestinian annotators despite Palestine

being a top-3 source country [Q31, Q89, WEB-3, DATASET-D4, DATASET-D5]. The 96% answer-key verification ceiling [Q43] likely reflects Jordanian/Egyptian consensus rather than pan-regional validity for contested historical and civic content. Strengths include authentic exam sourcing across eight countries, level-appropriate subject coverage, MSA text-only format matching the deployment modality, and confirmed substantial Palestinian STEM/curriculum content.

ID	Evidence
Q27	"Each question has 2–5 candidate answers, with one correct answer." (p.3)
Q31	"The data construction process involved a total of 10 Arabic native speakers from different countries: 6 internal workers (1 Jordanian, 1 Egyptian, 1 Lebanese, 1 from UAE, and 2 from KSA) and 4 external workers (3 Jordanian and 1 Egyptian)." (p.4)
Q43	"We found that 96% of the questions and answer keys match on average, while the remaining could have incorrect answer keys." (p.4)
Q88	"For future work, ArabicMMLU can be extended to include short-answer or essay questions, different modalities (i.e., images, audio, video), larger region coverage, and more questions in professional settings." (p.9)
Q89	"ArabicMMLU does not represent all Arabic countries equally. For example, we have collected over 6K multiple-choice questions from Jordan, while other countries are represented with only 100 questions or, in some cases, not at all." (p.9)
D4	لُدُع... يندرأل ابي برع ال شي ج ل تال ي ك ش ت ن م ة ي ب ر ع ال ة و ق ل ... 1952 ماع ندرأل ا ي ف ة ي ف ال ت ئ ا ة م و ك ح ل و ا ل ي ك ش ت م ت All 5 sampled Civics Middle School items are exclusively about Jordanian political history
D5	ي ن ط و ل ا ع ا ف د ل ا ب ة ف ل ك م ل ا ة ي م و ك ح ل ا ة ط ل س ل و ا ل د ع ل ا ر ي ز و ن م ... ف ا ن ئ ت س ا ل ا ل ج ا ئ د ت ب ي (Professional) items are from Morocco (Moroccan procedural law)
WEB-3	Palestine confirmed top-3 source country — yet no Palestinian annotators in pool (https://github.com/mbzuai-nlp/ArabicMMLU)

Practical Guidance

What This Benchmark Measures

ArabicMMLU measures a model's ability to select curriculum-designated correct answers on MCQ school-exam questions across 40 subjects spanning STEM, social science, humanities, Arabic language, and other domains, drawn predominantly from Jordan with substantial Egyptian and Palestinian content. The strongest dimension for this deployment is Input Ontology (subject coverage approximately matches the deployment's school-curriculum scope) and Input Form (uniform MSA text matches deployment modality). It can usefully assess whether a system has acquired the factual content of Jordanian-curriculum school knowledge in MSA.

Construct Depth

The benchmark probes factual recall and basic reasoning at primary, middle, and high school levels (university content only 6% [Q71]), but only at the depth of single-correct-answer MCQ. It does not probe explanatory quality, multi-perspective reasoning, perspective-flagging on contested content, robustness to learner-register Arabic, or appropriate framing for non-Muslim/non-Arab audiences — all of which are central to the deployment requirements. The Output Ontology and Output Form dimensions, scored 1, indicate that even a model achieving the 96% verification ceiling [Q44] would not have been evaluated on the pluralistic, explanatory behaviors the deployment demands.

What Else You Need

For valid deployment evaluation, substantial supplementation is required: (1) an open-ended Arabic generation evaluation such as AraGen with 3C3H LLM-as-judge [WEB-6] or 'Beyond MCQ' [WEB-5] to address the OO/OF gap; (2) a bespoke contested-content evaluation with Palestinian, Moroccan, and Kuwaiti subject-matter experts to address the OC gap; (3) Moroccan and Kuwaiti school-curriculum item collection to address the IO/IC coverage gaps for those deployment countries; (4) a learner-register Arabic robustness evaluation (none exists [WEB-6, gap_id 5]) to address the IF gap; and (5) explicit perspective-flagging evaluation specific to the eight-country contested-content scope, which no existing benchmark targets.

ID	Evidence
Q44	"This 96% score is considered to represent the maximum score meaningfully achievable for ArabicMMLU." (p.4)

Q71	"This could be attributed to the relatively small portion (i.e., 6%) of university-level questions in ArabicMMU, which potentially skews the results." (p.7)
WEB-5	'Beyond MCQ' Arabic open-ended benchmark — supplementary alternative for OF (https://arxiv.org/abs/2510.24328)
WEB-6	AraGen 3C3H LLM-as-judge framework for open-ended Arabic generation (https://arxiv.org/abs/2510.13430)